

Problem 10.4

You need \$500 on Jan 1, 2012. To save for this amount, you invest x on Jan 1, 2008 and $2x$ on July 1, 2008. The force of interest is $\delta_t = 0.02t$ where t is 0 on Jan 1, 2008. Find x .

Solution

The force of interest is

$$\delta_t = 0.02t.$$

The accumulation factor from time t_1 to time t_2 is

$$a(t_1, t_2) = e^{\int_{t_1}^{t_2} \delta_t dt}.$$

Substitute $\delta_t = 0.02t$:

$$a(t_1, t_2) = e^{\int_{t_1}^{t_2} 0.02t dt}.$$

Thus,

$$a(t_1, t_2) = e^{0.01(t_2^2 - t_1^2)}.$$

Jan 1, 2012 is time 4, and July 1, 2008 is time 0.5. Therefore, the accumulated value equation at time 4 is

$$xa(0, 4) + 2xa(0.5, 4) = 500.$$

Using the accumulation factor,

$$xe^{0.01(4^2 - 0^2)} + 2xe^{0.01(4^2 - 0.5^2)} = 500.$$

So

$$x(e^{0.16} + 2e^{0.1575}) = 500.$$

Hence,

$$x = \frac{500}{e^{0.16} + 2e^{0.1575}}.$$

Therefore,

$$x = 142.26077.$$

Thus,

$$\boxed{x = 142.26}.$$